Events and Sigma-Algebras

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1 Motivation

For a finite or countable sample space Ω we may assign probabilities to every subset. For uncountable Ω this fails: by the Vitali construction there exist subsets of $[0, 1]$ that admit no translation-invariant, countably-additive probability. To rescue the theory, we restrict attention to a distinguished family of *measurable* subsets called a sigma-algebra.

2 Definitions

Definition 1 (Sigma-algebra)**.** Let Ω be a non-empty set. A collection $\mathcal{F} \subseteq 2^\Omega$ is a *sigma-algebra* on Ω if

(S1) $\Omega \in \mathcal{F}$;

(S2) for every $A \in \mathcal{F}$, the complement $A^c = \Omega \setminus A$ also lies in $\mathcal{F}$;

(S3) for every sequence $A_1, A_2, \ldots \in \mathcal{F}$, the countable union $\bigcup_{n=1}^{\infty} A_n$ also lies in $\mathcal{F}$.

A pair $(\Omega, \mathcal{F})$ with $\mathcal{F}$ a sigma-algebra on Ω is a *measurable space*, and the elements of $\mathcal{F}$ are called *events* (or *measurable sets*).

Definition 2 (Generated sigma-algebra)**.** Let $\mathcal{C} \subseteq 2^\Omega$. The *sigma-algebra generated by* $\mathcal{C}$, written $\sigma(\mathcal{C})$, is the smallest sigma-algebra on Ω that contains $\mathcal{C}$, i.e.

$$\sigma(\mathcal{C}) \;=\; \bigcap \big\{\, \mathcal{F} \subseteq 2^\Omega \;:\; \mathcal{F} \text{ is a sigma-algebra on } \Omega,\ \mathcal{C} \subseteq \mathcal{F} \,\big\}.$$

The intersection is well-defined because 2^Ω is itself a sigma-algebra containing $\mathcal{C}$, and because of Theorem 4 below.

Definition 3 (Borel sigma-algebra)**.** The *Borel sigma-algebra* on $\mathbb{R}$ is $\mathcal{B}(\mathbb{R}) := \sigma(\mathcal{T})$, where $\mathcal{T}$ is the family of open subsets of $\mathbb{R}$. Equivalently, $\mathcal{B}(\mathbb{R}) = \sigma(\{(-\infty, a] : a \in \mathbb{R}\})$.

3 Key theorem

Theorem 4 (Arbitrary intersection of sigma-algebras)**.** *Let Ω be a non-empty set and let $\{\mathcal{F}_\alpha\}_{\alpha \in I}$ be a (possibly uncountable) family of sigma-algebras on Ω. Then*

$$\mathcal{G} \;:=\; \bigcap_{\alpha \in I} \mathcal{F}_\alpha$$

is a sigma-algebra on Ω.

Proof. We verify each of the three axioms (S1), (S2), (S3) for $\mathcal{G}$.

(S1) Whole space. For every $\alpha \in I$, axiom (S1) applied to $\mathcal{F}_\alpha$ gives $\Omega \in \mathcal{F}_\alpha$. Hence Ω lies in every member of the family, so

$$\Omega \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha = \mathcal{G}.$$

(S2) Closure under complement. Let $A \in \mathcal{G}$. By definition of intersection, $A \in \mathcal{F}_\alpha$ for every $\alpha \in I$. Applying axiom (S2) inside each $\mathcal{F}_\alpha$, we get $A^c \in \mathcal{F}_\alpha$ for every α. Therefore

$$A^c \in \bigcap_{\alpha \in I} \mathcal{F}_\alpha = \mathcal{G}.$$

(S3) Closure under countable union. Let $A_1, A_2, \ldots$ be a sequence in $\mathcal{G}$. Fix any $\alpha \in I$. Since each $A_n \in \mathcal{G} \subseteq \mathcal{F}_\alpha$, the sequence $\{A_n\}_{n \geq 1}$ lies entirely in $\mathcal{F}_\alpha$. By axiom (S3) applied to $\mathcal{F}_\alpha$,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_\alpha.$$

This holds for every $\alpha \in I$, so $\bigcup_{n=1}^{\infty} A_n \in \bigcap_\alpha \mathcal{F}_\alpha = \mathcal{G}$.

All three axioms hold; therefore $\mathcal{G}$ is a sigma-algebra on Ω. $\square$

Corollary 5. *For every class $\mathcal{C} \subseteq 2^\Omega$, the generated sigma-algebra $\sigma(\mathcal{C})$ exists, is unique, and equals the intersection of all sigma-algebras on Ω that contain $\mathcal{C}$.*

Proof. The family $\mathcal{S} := \{\mathcal{F} \subseteq 2^\Omega : \mathcal{F} \text{ is a sigma-algebra, } \mathcal{C} \subseteq \mathcal{F}\}$ is non-empty since $2^\Omega \in \mathcal{S}$. By Theorem 4, $\bigcap_{\mathcal{F} \in \mathcal{S}} \mathcal{F}$ is a sigma-algebra. It contains $\mathcal{C}$ by construction, and is the smallest such by minimality of intersection. $\square$

4 Examples

Example 6 (Trivial sigma-algebra)**.** $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is a sigma-algebra: $\Omega \in \mathcal{F}_0$; $\Omega^c = \emptyset \in \mathcal{F}_0$; any countable union of $\emptyset$'s and Ω's is either $\emptyset$ or Ω.

Example 7 (Sigma-algebra generated by one event)**.** For $A \subseteq \Omega$ with $A \notin \{\emptyset, \Omega\}$,

$$\sigma(\{A\}) = \{\emptyset, A, A^c, \Omega\}.$$